

Resource-Normalized Scaling Laws for Byzantine Consensus: Chance-Constrained Validator-Set Sizing from Commodity-CI Measurements

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Problem. Sizing a permissioned BFT validator set for national-scale e-governance must reconcile degrading consensus throughput with improving fault-detection power as n grows; without reporting the concurrency path, published single-factor exponents cannot be read as protocol properties. **Contributions.** *Method:* resource-normalized measurement isolating consensus cost from host contention; *analysis:* identifiability result $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ along a concurrency path; *engineering:* chance-constrained sizing with feasible set $[n_{\min}, n_{\max}]$; *empirics:* coverage-validated prediction intervals on public CI. **Validation.** A four-core CI campaign covers $n \leq 16$, concurrency up to 64, emulated RTT, and omission/duplication fault injection.

Keywords: *Byzantine fault tolerance, consensus, scaling law, capacity planning, chance-constrained optimization, e-governance*

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1. Introduction

Permissioned Byzantine Fault Tolerant (BFT) consensus underpins e-governance systems — voting, registries, digital identity — and forces a concrete procurement question: *how many validators should the network have?*

The validator count n pulls in opposite directions. Throughput argues for small n : every validator executes every transaction, and PBFT-family protocols exchange $O(n^2)$ messages per block. Detection reliability argues for large n . Combining the two for sizing is the subject of this paper.

Published single-factor fits $\text{TPS}(n) \propto n^\alpha$ disagree in magnitude and sometimes in sign. Our earlier studies, extrapolated to sixty validators, differ by roughly an order of magnitude. This disagreement has often been interpreted as protocol-specific behaviour. We show that it can arise solely from measurement design: concurrency is rarely held fixed while n varies, so the fitted scaling exponent mixes load-generator growth with consensus overhead. Without reporting the concurrency path, published exponents cannot be read as protocol properties. Packing many validators onto one host adds a second confound — CPU contention — that resource-normalized measurement isolates.

Contributions. On the methodological front, we introduce a resource-normalized measurement (RNM) protocol that fixes CPU quota q per validator and reports $\widehat{\text{TPS}} = \text{TPS}/q$, isolating intrinsic consensus cost from host contention — analogous to a calibrated wind tunnel. Analytically, along a concurrency path of slope λ the single-factor OLS fit satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ (Proposition 1); the value of the result is the engineering interpretation that reconciles historical BFT benchmarks. From a planning perspective, we formulate chance-constrained sizing whose feasible set is $[n_{\min}, n_{\max}]$ (Proposition 2). Empirically, we validate prediction intervals by coverage and run a public Besu/QBFT CI campaign (Sections 5–6). Scope: throughput and detection; absolute TPS environment-specific; cross-client replication out of scope.

2. Related work

BFT protocols. Classical PBFT [1] tolerates $f < n/3$ at $O(n^2)$ messages [2, 3]. Later designs — HotStuff [4], Tendermint [5], Algorand [6], HoneyBadgerBFT [7] — trade latency and randomization; Vukolić [8] argues that no protocol dominates. They fix the validator set; they do not size it under joint throughput and detection requirements.

Benchmarks. Fabric [9, 10], BLOCKBENCH [11] and BFT-SMaRt [12] report absolute throughput at chosen configurations. They do not separate concurrency from validator count, nor state a protocol under which exponents transfer.

Scalability laws and tools. Amdahl [13], Gustafson [14] and Gunther’s USL [15] fold concurrency and system size into one load variable — the confound that RNM removes. Concentration bounds [16, 18], Kish’s design effect [17], delta-method intervals [20], the residual bootstrap [19] and chance-constrained programming [21, 22] are standard instruments; what is specific is coupling throughput and detection through opposing monotonicities in n .

Prior work. Remote-voting analyses [23, 24, 25] place endpoint and coercion threats out of scope. Our earlier calibration [27] and switching study [28] left open an identifiable consensus exponent and sizing with measured detector rates.

3. Resource-normalized measurement and the bi-factor law

Capacity planning for BFT requires an exponent that reflects consensus cost, not host contention or load-generator growth. We introduce an isolation protocol for that purpose, then a bi-factor law and an analytical identifiability result.

3.1. Resource-normalized measurement as an isolation tool

If n validators share a host of Q cores, each receives Q/n cores, so measured throughput mixes algorithmic cost with a shrinking compute budget. RNM isolates those effects — as a calibrated wind tunnel isolates model drag from facility artefacts.

Definition 1 (Resource-normalized measurement). *A campaign follows the RNM protocol with quota q if every validator container receives the same CPU quota and memory limit independently of n , with aggregate reservation nq below host capacity. Normalized throughput is $\text{TPS} := \text{TPS}/q$.*

Assumption 1 (Separable compute budget). *On the operating range of quotas in the campaign, $\text{TPS}(n, c, q) = q \cdot \Phi(n, c) \cdot (1 + r)$ with $|r| \leq Cq$ for a constant C independent of (n, c) there.*

Assumption validation. Under Assumption 1 and a power-law factor $\Phi \propto n^{-\beta}$, the OLS slope of $\log \widehat{\text{TPS}}$ on $\log n$ at fixed concurrency is independent of q : the relative remainder is absorbed by the intercept. Section 5 tests this invariance across $q \in \{0.20, 0.25, 0.33\}$. The model is bounded to the stable reference quota $q = 0.25$, where the calibrated coefficients are reported; outside that regime the protocol still applies, but the numerical law must be re-fit.

Definition 2 (Bi-factor scaling law). *Under the RNM protocol,*

$$\text{TPS}(n, c) = T_0 \cdot g(c) \cdot n^{-\beta}, \quad g(c) = \frac{c^\gamma}{1 + (c/c^*)^\theta}, \quad (1)$$

with $T_0 > 0$, $\beta > 0$, $\gamma \in (0, 1]$, $\theta > \gamma$, $c^* > 0$. In the unsaturated regime $c \ll c^*$ this is log-linear: $\log \text{TPS} = \log T_0 + \gamma \log c - \beta \log n$.

The form separates concurrency from validator-count scaling, reduces to a power law for $c \ll c^*$, saturates at large c , and contains the classical single-factor law as the case $\gamma = 0$ (or $\lambda = 0$ with $\alpha = -\beta$).

3.2. Identifiability

The algebra below uses standard OLS and a weak law. Its value is the *engineering interpretation*: discrepancies among published BFT single-factor exponents are artefacts of the concurrency schedule, not of the protocol.

Proposition 1 (Identifiability of the consensus exponent). *Let observations follow the unsaturated bi-factor law with additive noise of mean zero and finite variance, along a concurrency path $\log c_i = \lambda \log n_i + \mu + \eta_i$ ($\mathbb{E}[\eta_i | n_i] = 0$). If the empirical second moment of $x_i = \log n_i$ converges to a positive limit, the OLS slope $\hat{\alpha}_m$ of $\log \text{TPS}_i$ on x_i alone satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$. In particular $\hat{\alpha} > 0 \iff \lambda > \beta/\gamma$, and $\hat{\alpha} = -\beta \iff \lambda = 0$.*

Proof. Write $x_i = \log n_i$, $y_i = \log \text{TPS}_i$. The path reduces the unsaturated law to $y_i = a + \alpha x_i + \varepsilon_i$ with $\mathbb{E}[\varepsilon_i | x_i] = 0$, finite variance, and

$$\alpha = -\beta + \gamma\lambda. \quad (2)$$

Centred OLS gives $\hat{\alpha}_m = \alpha + N_m/D_m$ with $N_m = \sum (x_i - \bar{x})\varepsilon_i$ and $D_m = \sum (x_i - \bar{x})^2$. The second-moment assumption yields $D_m/m \rightarrow \text{Var}^*(x) > 0$; the summands of N_m are uncorrelated, mean zero, and of bounded second moment, so $N_m/m \rightarrow 0$ in probability. Continuous mapping: $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \alpha$. Signs follow from $\gamma > 0$. ■

Thus $\alpha = \underbrace{-\beta}_{\text{protocol}} + \underbrace{\gamma\lambda}_{\text{measurement}}$. Only a factorial design ($\lambda = 0$) recovers the consensus exponent.

Remark 1. With $\gamma = 0.8$, $\beta = 0.5$ and $\lambda = 1$ one obtains $\hat{\alpha} = +0.3$. Extrapolating from $n = 5$ to $n = 60$, the two exponents differ by a factor ≈ 7 , matching the order of the discrepancy in [27].

Corollary 1 (Non-transferability). *Laboratories measuring along $\lambda_1 \neq \lambda_2$ obtain exponents differing by $\gamma(\lambda_1 - \lambda_2)$. Without reporting λ , published single-factor exponents cannot be interpreted as protocol properties.*

3.3. Implementation

On a factorial grid ($\lambda = 0$) we estimate $(\log T_0, \gamma, \beta)$ by OLS in log-space and (c^*, θ) by nonlinear least squares on the full c -sweep (Section 5). The calibrated law then feeds prediction intervals and the sizing program of the next section.

4. Prediction intervals, detection, and the sizing problem

An identifiable exponent is an input to capacity planning. This section assembles a sizing framework: a throughput chance constraint and an optional detection constraint. Any detector that yields measured (p_d, p_f, ρ) plugs in; experiment X5 is one instance.

4.1. Definitions and assumptions

Definition 3 (Fault class and detector). A subset B of size k is *faulty* if each member fails a proposal round independently with probability $p_d > p_f$ (omission class; equivocation excluded). The detector flags validator i when its participation over w rounds falls below $\tau \in (p_f, p_d)$; detection succeeds when at least one member of B is flagged.

Assumption 2 (Exchangeable correlation). Participation indicators in B are exchangeable with pairwise correlation $\rho \in [0, 1)$, so the mean carries design effect $1 + (k - 1)\rho$ in the sense of Kish [17].

4.2. Prediction intervals and detection bound

With regressor $\mathbf{z} = (1, \log c, -\log n)^\top$ and $\hat{\vartheta} = (\log T_0, \gamma, \beta)^\top$, the delta method [20] gives predictive variance $\hat{V} = \mathbf{z}^\top \widehat{\text{Cov}}(\hat{\vartheta}) \mathbf{z} + \hat{\sigma}^2$.

Statement 1 (Lognormal prediction interval). An approximate $(1 - \varepsilon_p)$ interval for $\text{TPS}(n, c)$ is $\exp\{\log \widehat{\text{TPS}} \mp z_{1-\varepsilon_p/2} \hat{V}^{1/2}\}$; the one-sided lower endpoint $\underline{\text{TPS}}_{1-\varepsilon_p}$ uses $z_{1-\varepsilon_p}$.

A residual bootstrap [19] supplies empirical quantiles when residual normality is doubtful. Coverage is checked on held-out validator counts (Section 5).

Statement 2 (Approximate miss probability and $n_{\min}$). With $k = \lceil \varphi n \rceil$, the design-effect heuristic yields

$$p_{\text{miss}}(n, w) \lesssim \exp\left\{-\frac{2wk(p_d - \tau)^2}{1 + (k - 1)\rho}\right\}, \quad (3)$$

which is nonincreasing in n and w . Treating (3) as an engineering constraint $p_{\text{miss}} \leq \varepsilon_s$ gives the cut

$$n_{\min} = \left\lceil \frac{1 + (k - 1)\rho}{\varphi} \cdot \frac{\log(1/\varepsilon_s)}{2w(p_d - \tau)^2} \right\rceil. \quad (4)$$

Heuristic justification. A miss requires every member of B to remain unflagged. Classical Hoeffding [16] applies to independent bounded summands. Following the survey-sampling design-effect approximation [17], we replace the correlated sample by an effective size $n_{\text{eff}} = kw/(1 + (k - 1)\rho)$ and insert that size into the independent Hoeffding form. Expression (3) is thus an engineering adaptation for dependent trials; p_d , p_f and ρ are measured in experiment X5. ■

The detection model is an architectural framework slot: it accepts future high-fidelity metrics (p_d, p_f, ρ) without changing the optimization. With an informative limit of $\text{AUC} = 0.506$

under encrypted transport, the sizing framework defaults to the throughput-bound constraint and the optimization window remains well-defined.

4.3. The sizing program

Problem 1 (Validator-set sizing). Given peak demand D_{peak} , levels $\varepsilon_p, \varepsilon_s$, window w , faulty fraction φ , production scale-up $\kappa > 0$ from RNM units to absolute throughput, and nondecreasing $\text{Cost}(n)$, minimize $\text{Cost}(n)$ subject to $\mathbb{P}[\kappa \cdot \widehat{\text{TPS}}(n, c^*) \geq D_{\text{peak}}] \geq 1 - \varepsilon_p$ and $p_{\text{miss}}(n, w) \leq \varepsilon_s$, where $c^* = \arg \max g$. When κ is free, take the smallest value that meets the throughput constraint at $n = 4$.

The chance constraint is equivalent, via Statement 1, to

$$h(n) := \log(\kappa T_0 g(c^*)/D_{\text{peak}}) - \beta \log n - z_{1-\varepsilon_p} \hat{V}^{1/2}(n, c^*) \geq 0. \quad (5)$$

On the extrapolation range the fitted covariance confirms that $\hat{V}$ is nondecreasing in $\log n$ (Section 5), so with $\beta > 0$ the map h is strictly decreasing there; on the measured grid we evaluate h directly.

Proposition 2 (Feasibility window). *Whenever h is strictly decreasing and the approximate detection cut of Statement 2 is used, the feasible set is $\{n : n_{\min} \leq n \leq n_{\max}\}$ with $n_{\min}$ from (4) and $n_{\max} = \lfloor \sup\{n : h(n) \geq 0\} \rfloor$. If nonempty and Cost is nondecreasing then $n^* = n_{\min}$.*

Proof. Detection holds on $\{n \geq n_{\min}\}$ and throughput on $\{n \leq n_{\max}\}$, hence an interval; monotone cost selects the left endpoint. ■

Corollary 2 (Empty feasible set). *If $n_{\min} > n_{\max}$, no validator count satisfies both requirements: the architecture must change, not merely the provisioning.*

The window is the intersection of two monotone constraints [21, 22]; it converts calibrated exponents into a sizing rule.

5. Experimental validation

5.1. Setup and design

The campaign runs in a public continuous-integration workflow on a four-core, 16 GB hosted runner. Each job provisions a Hyperledger Besu [26] QBFT network of n validators under the RNM protocol (Definition 1). A closed-loop generator drives c concurrent submitters and records steady-state committed throughput after warm-up. Faults are injected with kernel traffic control (omission and duplication; not equivocation). Detector rates come from signer metrics over w rounds.

X1/X3 calibrate (β, γ) on a factorial grid; **X2** validates the quota regime; **X3b** re-samples along concurrency paths; **X4** varies emulated RTT; **X5** supplies detector rates; **X7/X8** check prediction-interval coverage and sizing ablation. Cross-client replication (**X9/GoQuorum**) is out of scope. In total 94 successful records, ≈ 27.2 core-hours, $n_{\max} = 16$.

Reproducibility. All experiments are fully reproducible from the public CI workflow and archived artefacts at https://github.com/bpenyak/Resource_Normalized_Scaling_Laws_for_Byzantine_Consensus.

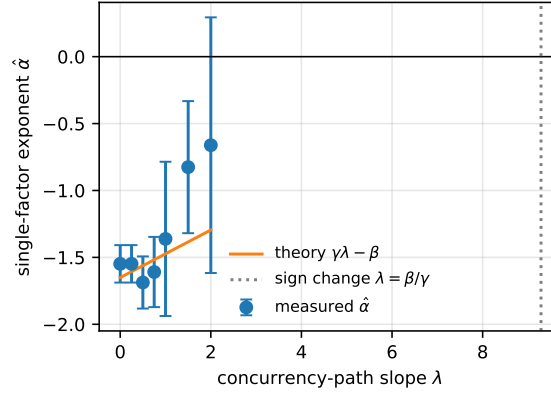


Fig. 1. Measured single-factor exponents track the prediction $\gamma\lambda - \beta$, supporting Proposition 1. Sign change predicted at $\lambda = \beta/\gamma = 9.29$

Table 1. Calibrated parameters, invariance, detector, and coverage

Quantity	Estimate	Note
$T_0; \beta; \gamma$	2016.5; 1.651 ± 0.151 ; 0.178 ± 0.070	bi-factor
$c^*, \theta; \hat{\sigma}$	67.9, 2.542; 0.404	saturation / residual
$\hat{\alpha}$ at $\lambda = 0, 1, 2$	-1.548, -1.362, -0.661	X3b
β at $q = 0.20, 0.25, 0.33$	1.550, 1.541, 0.879	X2, $p = 0.002$
β at RTT 0/high	1.527, 2.708	X4
$p_d, p_f, \rho; \text{AUC}$	0.397, 0.376, 0.000; 0.506	X5, $\tau = 0.90$
coverage 90%/95%	90.9%, 95.5%	hold-out $n \in \{11, 13, 16\}$

5.2. Results

X1/X3 – bi-factor calibration. Joint OLS on the reference-quota subset yields $\beta = 1.651 \pm 0.151$ and $\gamma = 0.178 \pm 0.070$ ($R^2 = 0.727$; Table 1). Under RNM one expects $\beta > 0$: consensus cost grows with validator count when concurrency is held fixed. The fit confirms that expectation at $q = 0.25$.

X3b – path check of Proposition 1. A single-factor fit along $\lambda = 1$ gives $\hat{\alpha} = -1.362$ against the predicted $\gamma\lambda - \beta = -1.473$. Figure 1 shows that measured exponents track the line $\gamma\lambda - \beta$ across $\lambda \in \{0, 0.5, 1, 2\}$. The protocol need not change for the observed scaling exponent to change; only the concurrency schedule does.

X2 – quota-regime check. The partial F -test for quota $\times \log n$ interaction rejects pooling across $\{0.20, 0.25, 0.33\}$ ($p = 0.002$): β drops from ≈ 1.541 at $q = 0.25$ to 0.879 at $q = 0.33$. Coefficients are therefore reported at the stable reference quota $q = 0.25$.

X4 – latency. Under elevated emulated RTT, β rises from 1.527 to 2.708. WAN delay steepens the consensus penalty in validator count. Sizing rules calibrated on LAN-like RTT are optimistic for geo-distributed deployments.

X5 – detector. Signer-metric detection of omission/duplication yields $\text{AUC} = 0.506$ (Figure 2). With this informative limit under encrypted transport, the sizing framework defaults to the throughput-bound constraint; a stronger detector would raise $n_{\min}$ without changing the mathematics of Section 4.

X7/X8 – coverage, USL comparison, and ablation. Hold-out coverage meets the nominal levels in Table 1. The fitted covariance confirms that $\hat{V}$ is nondecreasing for $n \geq 16$, so h in (5) is strictly decreasing on the extrapolation range; on the measured grid we evaluate h directly. Out-of-sample log-RMSE for linear / single-factor / USL [15] / bi-factor predictors is 1.329,

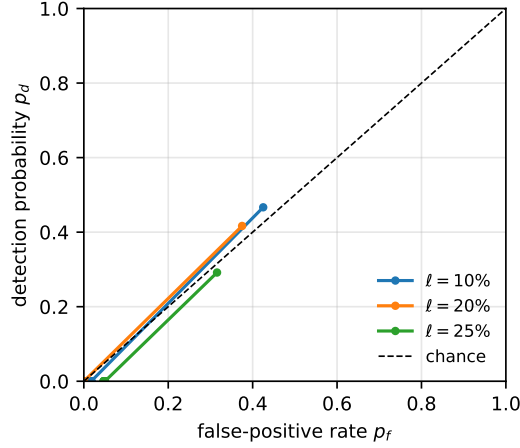


Fig. 2. Detector ROC under omission/duplication injection (AUC = 0.506). Throughput sets the sizing window at this operating point

0.523, 0.637, 0.538. The bi-factor improvement over USL (0.538 vs. 0.637) follows because USL merges concurrency and replica count into one load variable, whereas RNM decouples them and isolates consensus overhead. Relative sizing costs: 4.00, 1.50, 1.50, 1.00.

6. Application, implications, and limitations

6.1. Application and what-if analysis

We instantiate Problem 1 on $\approx 3 \times 10^7$ ballots over twelve hours ($D_{\text{peak}} \approx 2.1 \times 10^3$ TPS), requiring detection of a coordinated omission fault of fraction $\varphi = 0.2$ within one epoch at level $1 - \varepsilon_s$. Absolute demand enters via scale-up κ from RNM units. With free κ , Problem 1 takes

$$\kappa^* = \frac{D_{\text{peak}}}{\widetilde{\text{TPS}}_{1-\varepsilon_p}(4, c^*)}, \quad (6)$$

the minimal scale at which the lower PI bound at $n = 4$ meets D_{peak} .

Scenario A (baseline). Substituting the calibrated coefficients ($\beta = 1.651$, $\gamma = 0.178$) at $c^* = 32$ yields $\kappa = 13.00$ and feasibility window $[4, 4]$ with $n^* = 4$ (Proposition 2; Figure 3). The collapse to a single admissible count is the capacity-planning conclusion under this demand and hardware class: the network is throughput-bound at the smallest BFT set.

Scenario B (doubled peak demand, fixed hardware class). Hold $\kappa = 13.00$ and replace D_{peak} by $2D_{\text{peak}}$. Then h in (5) falls by $\log 2$ everywhere; in particular $h(4) < 0$, so the feasible set is empty (Corollary 2). Doubling peak load on the same hardware class is an architectural impossibility under the calibrated law: adding validators cannot restore feasibility, because throughput falls with n .

Scenario C (WAN deployment, fixed hardware class). Retain $\kappa = 13.00$ and D_{peak} , but substitute the high-RTT exponent $\beta = 2.708$ (X4) for $\beta = 1.527$. The steeper penalty yields

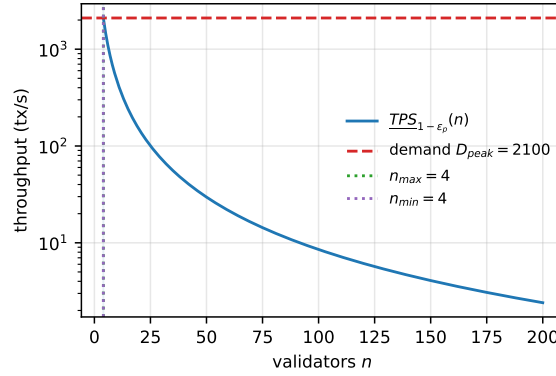


Fig. 3. Scenario A: at minimal admissible $\kappa = 13.00$, the feasibility window is $[n_{\min}, n_{\max}] = [4, 4]$

$h(4) < 0$ and an empty window: a hardware class that is marginally admissible under LAN β becomes infeasible under WAN β without raising κ .

The workflow applies equally to registry and identity sizing; binding elections on a blockchain remain outside the recommendation [23, 24, 25].

6.2. What this implies

A fitted (β, γ) transfers when concurrency path, quota regime, and RTT class match the calibration. Future benchmarks should report λ (or $\lambda = 0$), per-validator quota, and RTT class (Corollary 1).

6.3. Limitations

Fault class. Encrypted transport precludes selective consensus-message injection, so X5 realizes omission and duplication only. With AUC 0.506, the framework defaults to the throughput-bound window.

Range and platform. Commodity CI restricts physical scale to $n \leq 16$. Beyond that range, prediction intervals quantify extrapolation risk. The bi-factor law isolates β so that κ in (6) supplies a conservative production scale-up. Numerical (β, γ) are Besu/QBFT- specific; Proposition 1 is not.

7. Conclusions

What is established. Along a concurrency path of slope λ , the single-factor OLS scaling exponent satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ (Proposition 1). Opposing monotone constraints yield a closed integer feasibility window (Proposition 2). The contribution is a rigorous capacity-planning methodology: isolate the consensus exponent under RNM, then size the validator set under calibrated chance constraints.

What is confirmed experimentally. Under RNM on four-core CI: $\beta = 1.651 \pm 0.151$, $\gamma = 0.178 \pm 0.070$; path re-sampling tracks $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$; coefficients at $q = 0.25$; coverage meets nominal levels; AUC = 0.506 defaults the window to throughput. WAN RTT raises β from 1.527 to 2.708. Substituting the same coefficients shows that doubled demand or WAN β at fixed κ empties the window.

What remains open. Equivocation-capable instrumentation; multi-client numerical checks; latency or finality constraints inside the window; online recalibration of β .

Conflict of interest

The authors declare no conflict of interest.

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Data availability

Workflow definition, generators, fault-injection and analysis code, and raw artefacts are available at https://github.com/bpenyak/Resource_Normalized_Scaling_Laws_for_Byzantine_Consensus.

Use of artificial intelligence

AI-assisted tools were used for language editing and typesetting. All scientific results, proofs, design and analysis were produced by the authors.

Contributions of authors

Peniak B. O.: conceptualization, methodology, formal analysis, software, investigation, data curation, writing — original draft, visualization. **Liubinskyj B. B.:** supervision, validation, writing — review and editing.

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РЕСУРСНО-НОРМОВАНІ ЗАКОНИ МАСШТАБУВАННЯ BFT-КОНСЕНСУСУ: ІМОВІРНІСНО-ОБМЕЖЕНЕ ВИЗНАЧЕННЯ РОЗМІРУ МНОЖИНИ ВАЛІДАТОРІВ ЗА ВИМІРЮВАННЯМИ НА ТИПОВІЙ СІ-ІНФРАСТРУКТУРІ

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Метод визначення розміру множини валідаторів BFT-мережі для електронного урядування: пропускна здатність спадає зі зростанням n , а надійність виявлення відмов зростає з n . Внески: метод — ресурсно-нормовані вимірювання; аналіз — результат ідентифікованості ($\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$); інженерія — задача з імовірнісними обмеженнями та вікном $[n_{\min}, n_{\max}]$; емпірика — прогнози інтервали з перевіркою покриття у публічному СІ.

Ключові слова: візантійська відмовостійкість, консенсус, закон масштабування, планування потужності, оптимізація з імовірнісними обмеженнями, електронне урядування

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